# Excel Notes for Analyst Jobs

18 pages. Every formula shows its answer.
Tested before you read it.

# 1. CELL REFERENCES

**THE BASE OF EVERY FORMULA**

By @thedataguy16

1. *Price x quantity*

```
=A2*B2
```
→ 1000

2. *GST on every row ($B$1 fixed)*

```
=A2*$B$1
```
→ 45

3. *Copy right, row stays fixed*

```
=A2*B$1
```
→ 45

4. *Total of a column*

```
=SUM(B2:B6)
```
→ 150

5. *Average*

```
=AVERAGE(B2:B6)
```
→ 30

6. *Largest value*

```
=MAX(B2:B6)
```
→ 50

7. *How many numeric entries*

```
=COUNT(A2:A6)
```
→ 3

8. *How many rows are filled*

```
=COUNTA(A2:A6)
```
→ 4

9. *Round to 2 places*

```
=ROUND(A2,2)
```
→ 3.14

10. *Always round up*

```
=ROUNDUP(A2,0)
```
→ 4

# 2. IF LOGIC

LET EXCEL DECIDE

By @thedataguy16

1 *Pass / fail on marks*

```
=IF(B2>=40,"Pass","Fail")
```
→ Pass

2 *Grade with nested IF*

```
=IF(B2>=75,"A",IF(B2>=60,"B","C"))
```
→ B

3 *Same grade with IFS*

```
=IFS(B2>=75,"A",B2>=60,"B",TRUE,"C")
```
→ B

4 *Eligible for a campus drive*

```
=IF(AND(B2>=40,C2>=75),"Eligible","No")
```
→ Eligible

5 *Is the ticket done?*

```
=IF(OR(D2="Delivered",D2="Closed"),"Done","Open")
```
→ Done

6 *Is the cell empty?*

```
=IF(ISBLANK(E2),"Missing","Filled")
```
→ Missing

7 *Stop #DIV/0! in reports*

```
=IFERROR(A2/B2,0)
```
→ 0

8 *Missing in master list*

```
=IFNA(MATCH("Z",A2:A4,0),"Not found")
```
→ Not found

9 *Region code to name*

```
=SWITCH(F2,"N","North","S","South","Other")
```
→ South

# 3. SUM WITH CONDITIONS

**SUMIFS RUNS HALF OF MIS**

By @thedataguy16

Sample data: A = North, South, North, East, North, South | B = 100..600 | C = status

1. *Orders from North*

```
=COUNTIF(A2:A7,"North")
```
→ 3

2. *Count with 2 conditions*

```
=COUNTIFS(A2:A7,"North",C2:C7,"Closed")
```
→ 2

3. *Sum one region*

```
=SUMIF(A2:A7,"South",B2:B7)
```
→ 800

4. *North + Closed revenue*

```
=SUMIFS(B2:B7,A2:A7,"North",C2:C7,"Closed")
```
→ 400

5. *Sum values above 250*

```
=SUMIF(B2:B7,">250")
```
→ 1800

6. *Average one region*

```
=AVERAGEIF(A2:A7,"North",B2:B7)
```
→ 300

7. *Best South deal*

```
=MAXIFS(B2:B7,A2:A7,"South")
```
→ 600

8. *Smallest pending deal*

```
=MINIFS(B2:B7,C2:C7,"Open")
```
→ 200

9. *Count not closed*

```
=COUNTIF(C2:C7,"<>Closed")
```
→ 2

10. *Revenue = price x qty, all rows*

```
=SUMPRODUCT(B2:B7,D2:D7)
```
→ 2800

# 4. LOOKUPS

## VLOOKUP TO XLOOKUP

By @thedataguy16

Sample data: A2:A4 = 101, 102, 103 | B = Asha, Ravi, Neha | C = Pune, Indore, Jaipur | E2 = 102 | A5:B5 = 102, Ravi K | G2 = 72000, slab H2:I4 = 0 Low, 50000 Mid, 100000 High

1. *Name from employee ID*

```
=VLOOKUP(E2,A2:C4,2,FALSE)
```

→ Ravi

2. *City from employee ID*

```
=VLOOKUP(E2,A2:C4,3,0)
```

→ Indore

3. *Salary band from a slab table*

```
=VLOOKUP(G2,H2:I4,2,TRUE)
```

→ Mid

4. *Position in a list*

```
=MATCH(E2,A2:A4,0)
```

→ 2

5. *Value at a position*

```
=INDEX(B2:B4,2)
```

→ Ravi

6. *INDEX + MATCH, works anywhere*

```
=INDEX(B2:B4,MATCH(E2,A2:A4,0))
```

→ Ravi

7. *ID from a name (left lookup)*

```
=INDEX(A2:A4,MATCH("Neha",B2:B4,0))
```

→ 103

8. *XLOOKUP (Excel 365)*

```
=XLOOKUP(E2,A2:A4,B2:B4)
```

→ Ravi

9. *With a not-found value*

```
=XLOOKUP(999,A2:A4,B2:B4,"Not found")
```

→ Not found

10. *Last match, not first*

```
=XLOOKUP(E2,A2:A5,B2:B5,,0,-1)
```

→ Ravi K

# 5. CLEAN TEXT

## MESSY NAMES, FIXED

By @thedataguy16

1. *Fix pasted names*

   `=TRIM(A2)` → ravi KUMAR

2. *Clean name for an offer letter*

   `=PROPER(TRIM(A2))` → Ravi Kumar

3. *Capital letters*

   `=UPPER(B2)` → PUNE

4. *Small letters*

   `=LOWER(B2)` → pune

5. *Invoice prefix*

   `=LEFT(C2,3)` → INV

6. *Invoice number*

   `=RIGHT(C2,4)` → 2041

7. *From the middle*

   `=MID(C2,5,2)` → 20

8. *Length*

   `=LEN(C2)` → 8

9. *Position of a dash*

   `=FIND("-",C2)` → 4

10. *Clean phone numbers*

    `=SUBSTITUTE(D2,"-","")` → 9820011111

# 6. JOIN AND SPLIT

## NAMES, EMAILS, LABELS

By @thedataguy16

1 *Join with a space*

```
=A2&" "&B2
```
→ Asha Rao

2 *Same with CONCAT*

```
=CONCAT(A2," ",B2)
```
→ Asha Rao

3 *Join a list, skip blanks*

```
=TEXTJOIN(", ",TRUE,A2:A4)
```
→ Pune, Indore

4 *Username from email*

```
=LEFT(C2,FIND("@",C2)-1)
```
→ asha

5 *Company from email*

```
=MID(C2,FIND("@",C2)+1,50)
```
→ corp.in

6 *Number pasted as text*

```
=VALUE(E2)
```
→ 1250

7 *Percent label for a slide*

```
=TEXT(F2,"0.0%")
```
→ 25.6%

8 *Show a date your way*

```
=TEXT(G2,"dd-mmm-yyyy")
```
→ 05-Mar-2026

9 *Repeat a character*

```
=REPT("*",H2)
```
→ ****

# 7. DATES

## REPORTS RUN ON DATES

By @thedataguy16

Sample data: A2 = 15-Mar-2026 | B2 = 15-Oct-2026

1. *Today's date*

```
=TODAY()>DATE(2026,1,1)
```
→ TRUE

2. *Year*

```
=YEAR(A2)
```
→ 2026

3. *Month number*

```
=MONTH(A2)
```
→ 3

4. *Month label for a pivot*

```
=TEXT(A2,"mmm")
```
→ Mar

5. *Month-end for MIS cut-off*

```
=EOMONTH(A2,0)
```
→ 31-Mar-2026

6. *Probation end date*

```
=EDATE(A2,3)
```
→ 15-Jun-2026

7. *Months of experience*

```
=DATEDIF(A2,B2,"m")
```
→ 7

8. *Working days to close a ticket*

```
=NETWORKDAYS(A2,B2)
```
→ 154

9. *Weekday (Mon = 1)*

```
=WEEKDAY(A2,2)
```
→ 7

10. *30 days later*

```
=A2+30
```
→ 14-Apr-2026

# 8. EXCEL 365 FORMULAS

**NOT IN EXCEL 2016 OR 2019**

By @thedataguy16

Sample data: A = Pune, Indore, Pune, Jaipur | B = Asha, Ravi, Neha, Imran | C = 300, 900, 700, 200

1. *Each city once*

```
=UNIQUE(A2:A5)
```

→ Pune, Indore, Jaipur

2. *Cities seen only once*

```
=UNIQUE(A2:A5,,TRUE)
```

→ Indore, Jaipur

3. *Count distinct cities*

```
=ROWS(UNIQUE(A2:A5))
```

→ 3

4. *Sort by sales, high first*

```
=SORTBY(A2:C5,C2:C5,-1)
```

→ Indore 900, Pune 700...

5. *Rows with sales over 500*

```
=FILTER(A2:C5,C2:C5>500)
```

→ Indore 900, Pune 700

6. *Same, with a fallback*

```
=FILTER(A2:C5,C2:C5>5000,"None")
```

→ None

7. *Two conditions (AND)*

```
=FILTER(A2:C5,(A2:A5="Pune")*(C2:C5>500))
```

→ Pune, Neha, 700

8. *Split at commas*

```
=TEXTSPLIT(D2,",")
```

→ Pune | Indore | Jaipur

9. *Split, skip empty bits*

```
=TEXTSPLIT(D2,",",,TRUE)
```

→ Pune | Indore | Jaipur

# 9. PERCENT AND GROWTH

**EVERY MIS REVIEW ASKS THESE**

By @thedataguy16

Sample data: B2:B5 = 100, 200, 300, 400 growth rows use B2 = last month, C2 = this month

1 *Region share of total*

`=B2/SUM($B$2:$B$5)` → 10%

2 *Growth vs last month*

`=(C2-B2)/B2` → 25%

3 *Running total*

`=SUM($B$2:B4)` → 600

4 *Rank the sales team*

`=RANK.EQ(B3,$B$2:$B$5)` → 3

5 *Second highest*

`=LARGE(B2:B5,2)` → 300

6 *Lowest*

`=SMALL(B2:B5,1)` → 100

7 *Average yearly growth (CAGR)*

`=(C2/B2)^(1/3)-1` → 10%

8 *Percent as a label*

`=ROUND(B2/C2*100,1)&"%"` → 33.3%

9 *Growth when last month was 0*

`=IFERROR((C2-B2)/B2,"")` → (blank)

# 10. SPOT DIRTY DATA

**CHECK BEFORE YOU ANALYSE**

By @thedataguy16

1. *Duplicate invoice?*

```
=COUNTIF($A$2:$A$6,A2)>1
```
→ TRUE

2. *Label duplicates*

```
=IF(COUNTIF($A$2:$A$6,A2)>1,"Duplicate","")
```
→ Duplicate

3. *Is it blank?*

```
=ISBLANK(B2)
```
→ TRUE

4. *Is it a real number?*

```
=ISNUMBER(C2)
```
→ FALSE

5. *Number with spaces around it*

```
=VALUE(TRIM(C2))
```
→ 125

6. *Remove line breaks*

```
=CLEAN(D2)
```
→ PuneWest

7. *Count blank cells*

```
=COUNTBLANK(B2:B6)
```
→ 2

8. *Extra spaces hiding*

```
=LEN(F2)-LEN(TRIM(F2))
```
→ 2

9. *Exact match incl. case*

```
=EXACT(G2,H2)
```
→ FALSE

# 11. Pivot tables, step by step

by @thedataguy16

| # | Step | How |
|---|---|---|
| 1 | Turn data into a table | `Click inside the data, Insert > Table` |
| 2 | Insert the pivot | `Insert > PivotTable > New Worksheet` |
| 3 | Sales by region | `Drag Region to Rows, Sales to Values` |
| 4 | Split by month | `Drag Month to Columns` |
| 5 | Count instead of sum | `Value Field Settings > Count` |
| 6 | Show % of total | `Show Values As > % of Grand Total` |
| 7 | Group dates by month | `Right-click a date > Group > Months` |
| 8 | Filter one region | `Drag Region to Filters` |
| 9 | Clickable filter buttons | `PivotTable Analyze > Insert Slicer` |
| 10 | Biggest first | `Right-click a value > Sort > Largest to Smallest` |
| 11 | Show only the best 5 | `Row label filter > Value Filters, set it to 5` |
| 12 | New data added | `Right-click the pivot > Refresh` |
| 13 | Blanks should show 0 | `PivotTable Options > For empty cells show: 0` |
| 14 | Make a chart from it | `PivotTable Analyze > PivotChart` |
| 15 | See rows behind a number | `Double-click that number` |
| 16 | Flat, report-style layout | `Design > Report Layout > Show in Tabular Form` |

# 12. CONDITIONAL FORMATTING

## RULES THAT USE A FORMULA

By @thedataguy16

1. Sales above 1000

```
=$C2>1000
```

→ TRUE

2. Pending orders

```
=$D2="Pending"
```

→ TRUE

3. Duplicates in column A

```
=COUNTIF($A:$A,$A2)>1
```

→ TRUE

4. Highest value

```
=C2=MAX($C$2:$C$10)
```

→ TRUE

5. Every other row

```
=MOD(ROW(),2)=0
```

→ TRUE

6. Overdue dates

```
=$E2<TODAY()
```

→ TRUE

7. Empty cells

```
=ISBLANK($B2)
```

→ TRUE

8. Between 500 and 999

```
=AND($C2>=500,$C2<1000)
```

→ TRUE

9. Weekend dates

```
=WEEKDAY($E2,2)>5
```

→ TRUE

10. Text contains delhi

```
=ISNUMBER(SEARCH("delhi",$F2))
```

→ TRUE

# 13. MIS REPORT FORMULAS

**MONTHLY REVIEW, SORTED**

By @thedataguy16

Sample data: A = region | B = Jan, Jan, Feb, Mar | C = 100, 200, 300, 400

1 *January total*

`=SUMIFS(C:C,B:B,"Jan")` → 300

2 *North in January*

`=SUMIFS(C:C,A:A,"North",B:B,"Jan")` → 100

3 *Orders in a month*

`=COUNTIFS(B:B,"Mar")` → 1

4 *Share of one region*

`=SUMIF(A:A,"North",C:C)/SUM(C:C)` → 80%

5 *Target met?*

`=IF(D2>=E2,"Met","Missed")` → Met

6 *Target achievement*

`=D2/E2` → 120%

7 *Month on month*

`=D3/D2-1` → 20%

8 *Weighted average price*

`=SUMPRODUCT(F2:F4,G2:G4)/SUM(G2:G4)` → 22.5

9 *Year to date*

`=SUM($D$2:D4)` → 350

10 *Gap to target*

`=MAX(0,E2-D2)` → 20

# 14. Which chart, when

by @thedataguy16

| # | You want to show | Use |
|---|---|---|
| 1 | Change over months | `Line chart` |
| 2 | Compare a few categories | `Column chart` |
| 3 | Long category names | `Horizontal bar chart` |
| 4 | Rank many items | `Bar chart, sorted` |
| 5 | Share of a whole, 2 to 4 parts | `Pie or doughnut` |
| 6 | Share, month by month | `Stacked column` |
| 7 | Two measures, different scales | `Combo chart with a secondary axis` |
| 8 | Link between two numbers | `Scatter chart` |
| 9 | How values are spread | `Histogram` |
| 10 | Ups and downs to a final total | `Waterfall chart` |
| 11 | One headline number | `A big number with a label` |
| 12 | Actual vs target | `Column with a target line` |
| 13 | Few causes, most of the effect | `Pareto chart` |
| 14 | Sales by state | `Map chart (Excel 365)` |

# 15. Shortcuts that save time

by @thedataguy16

| # | Do this | Keys (Windows) |
|---|---|---|
| 1 | Make a table | `Ctrl + T` |
| 2 | Filters on or off | `Ctrl + Shift + L` |
| 3 | AutoSum | `Alt + =` |
| 4 | Flash Fill | `Ctrl + E` |
| 5 | Quick chart | `Alt + F1` |
| 6 | Chart on a new sheet | `F11` |
| 7 | Format Cells | `Ctrl + 1` |
| 8 | Add $ to a reference | `F4` |
| 9 | Fill down | `Ctrl + D` |
| 10 | Edit the cell | `F2` |
| 11 | New line inside a cell | `Alt + Enter` |
| 12 | Select the column | `Ctrl + Space` |
| 13 | Select the row | `Shift + Space` |
| 14 | Insert today's date | `Ctrl + ;` |
| 15 | Paste Special | `Ctrl + Alt + V` |
| 16 | Jump to the edge of data | `Ctrl + Arrow key` |

# 16. Errors and how to fix them

by @thedataguy16

| # | Error | What it usually means |
|---|---|---|
| 1 | #N/A | Lookup value not found. Check spaces, text vs number. |
| 2 | #VALUE! | Text where a number was expected. |
| 3 | #REF! | The formula points to a deleted cell. |
| 4 | #DIV/0! | Dividing by zero or a blank cell. |
| 5 | #NAME? | Misspelt function, missing quotes, or not in your version. |
| 6 | #SPILL! | A 365 formula has no room. Clear the cells below. |
| 7 | #NUM! | An impossible number, like a bad date sum. |
| 8 | ##### | Column too narrow. Widen it. |
| 9 | Total looks wrong | Numbers stored as text. They sit on the left. |
| 10 | VLOOKUP picks a wrong row | Last argument missing. Use FALSE. |
| 11 | Formula not updating | Formulas > Calculation Options > Automatic. |
| 12 | Circular reference | The formula uses its own cell. |

# 17. Interview questions, short answers

by @thedataguy16

| # | Question | Answer |
|---|---|---|
| 1 | VLOOKUP vs XLOOKUP? | XLOOKUP can look left, has a not-found value and is exact by default. |
| 2 | What does $ do in $A$1? | Locks the column and row when you copy. |
| 3 | COUNT vs COUNTA? | COUNT: numbers only. COUNTA: any filled cell. |
| 4 | Remove duplicates? | Data > Remove Duplicates, or UNIQUE in 365. |
| 5 | SUMIF vs SUMIFS? | SUMIF: one condition. SUMIFS: many, sum range first. |
| 6 | What is a pivot table? | A drag-and-drop summary. No formulas needed. |
| 7 | Find duplicates, keep data? | COUNTIF(range, cell)>1 or highlight rules. |
| 8 | What is Flash Fill? | Ctrl + E. Excel repeats the pattern you typed. |
| 9 | Why INDEX + MATCH? | Looks both ways and survives inserted columns. |
| 10 | How do you handle errors? | Fix the cause, then IFERROR or IFNA. |
| 11 | Relative vs absolute? | Relative moves when copied. Absolute stays. |
| 12 | What is data validation? | Rules for what a cell accepts, like a drop-down. |
| 13 | Freeze the header row? | View > Freeze Panes > Freeze Top Row. |
| 14 | What is Power Query for? | Repeatable cleaning: import, clean, combine, refresh. |

# 18. Practise these 12 tasks

by @thedataguy16

| # | Task | You will use |
|---|---|---|
| 1 | Total sales by region and month | `SUMIFS or a pivot` |
| 2 | Fill a customer's city from another sheet | `XLOOKUP or INDEX + MATCH` |
| 3 | Clean 500 messy names | `TRIM, PROPER` |
| 4 | Split emails into name and domain | `LEFT, MID, FIND` |
| 5 | Flag duplicate invoices | `COUNTIF` |
| 6 | Grade students into A/B/C | `IFS` |
| 7 | Month-on-month growth table | `Percent formulas` |
| 8 | Overdue payments in red | `Conditional formatting` |
| 9 | Best 5 products by revenue | `SORTBY or a sorted pivot` |
| 10 | Working days per order | `NETWORKDAYS` |
| 11 | One-page sales dashboard | `Pivot + slicer + chart` |
| 12 | Target vs actual report | `IF, percent, column chart` |

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